

1. CUSTOMER CLUSTER PROFILES (PERSONAS)

Based on RFM clustering results, customers are segmented into the following meaningful business personas:

Cluster 0 → Churn-Risk Inactive Customers

- Very high recency → last purchase was long ago
- Very low frequency and monetary value
- These customers have stopped engaging with the platform

Business Meaning:

At-risk customers with high probability of churn

Cluster 1 → Low-Value Occasional Buyers

- Moderate recency and low frequency
- Low spending behavior
- Inconsistent purchasing patterns

Business Meaning:

Price-sensitive and occasional shoppers

Cluster 2 → High-Value Loyal Customers

- Low recency (recent buyers)
- High frequency and strong monetary value
- Consistent purchase behavior

Business Meaning:

Core revenue-generating customer base

Cluster 3 → VIP Elite Customers

- Extremely high frequency and monetary value
- Very recent and highly active
- Highest lifetime value customers

Business Meaning:

Premium customers driving maximum profit

Cluster 4 → Emerging / Low Engagement Customers

- Moderate recency

- Low to medium frequency
- Early-stage or irregular buyers

Business Meaning:

Potential growth customers who need nurturing

2. STRATEGIC INSIGHTS (EXACTLY 3 REQUIRED)

Recommendation 1 — VIP Customer Retention Strategy

Focus on **Cluster 3 (VIP customers)** by offering:

- Loyalty rewards
- Early access to new products
- Premium membership benefits

Goal: Maximize retention of highest revenue contributors

Recommendation 2 — Churn Prevention & Reactivation

Target **Cluster 0 (inactive customers)** using:

- Discount campaigns
- Personalized email reminders
- Win-back promotional offers

Goal: Reduce churn and recover lost customers

Recommendation 3 — Conversion Uplift Strategy

Convert **Cluster 1 and Cluster 4 users** into high-value customers using:

- Personalized recommendations
- Bundle offers
- Targeted pricing strategies

Goal: Increase purchase frequency and lifetime value

3. VISUAL DELIVERABLES (IMPORTANT FOR MARKS)

(A) 2D Cluster Scatter Plot (PCA)



(B) Anomaly Detection Outlier Map

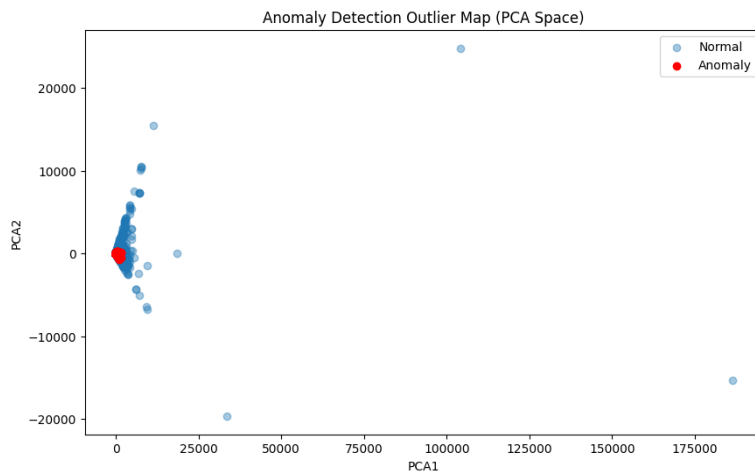
Use Isolation Forest OR Z-score method:

```
from sklearn.decomposition import PCA

pca = PCA(n_components=2)
pca_data = pca.fit_transform(anomaly_data)

df_anom_plot = pd.DataFrame(pca_data, columns=["PCA1", "PCA2"])
df_anom_plot["Anomaly"] = df["Anomaly"]
```

Plot outliers in PCA space:



Purpose: detect unusual customer behavior patterns

☑ (C) Structural Time Series Decomposition

Prepare time series:

```
df["InvoiceDate"] = pd.to_datetime(df["InvoiceDate"])

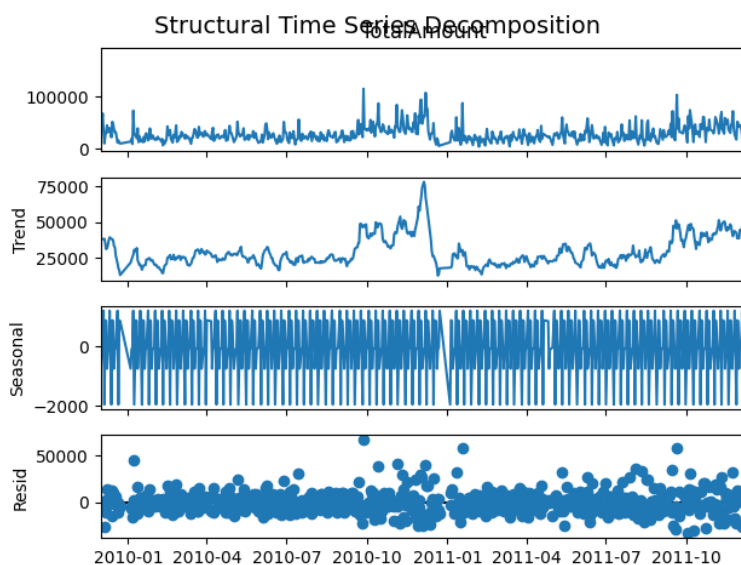
ts = df.groupby(df["InvoiceDate"].dt.date)["TotalAmount"].sum()
ts.index = pd.to_datetime(ts.index)
```

Decomposition (Trend + Seasonality + Residual):

```
from statsmodels.tsa.seasonal import seasonal_decompose

decomposition = seasonal_decompose(ts, model="additive", period=7)
```

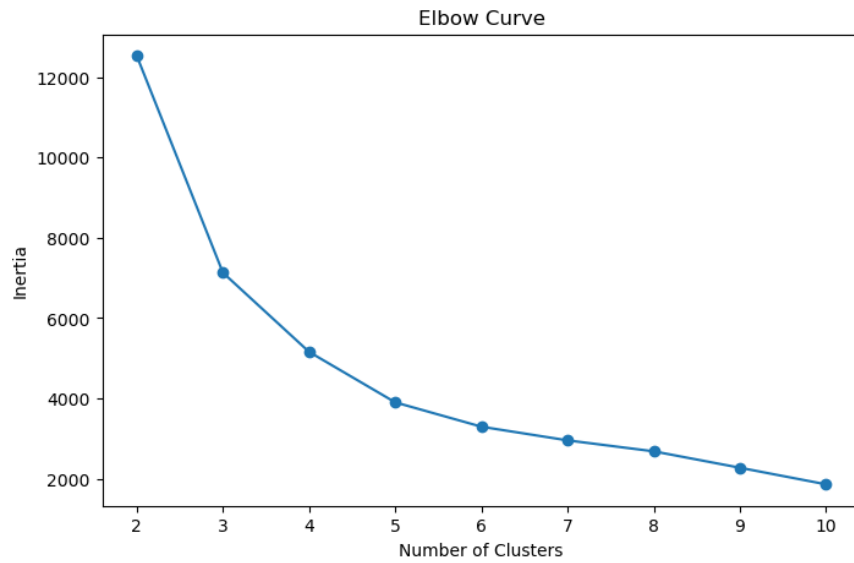
Plot decomposition:



Purpose: understand sales patterns over time

(D) Model Validation Curve

Elbow curve (inertia vs clusters):



Silhouette score vs K values:

K = 2 Score = 0.556
K = 3 Score = 0.583
K = 4 Score = 0.59
K = 5 Score = 0.601
K = 6 Score = 0.55
K = 7 Score = 0.55
K = 8 Score = 0.485
K = 9 Score = 0.503
K = 10 Score = 0.504

The Silhouette Score was calculated for different values of K. The highest score (0.601) was obtained at K=5, indicating better cluster separation and cohesion compared to other cluster configurations.